Software Test Plan (STP) - Web Based ML Model Evaluator

Project: Web based ML Model Evaluator
Version: 1.0
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Date: 17-09-2025
Status: Sample / Draft

1. Introduction

Purpose: This document defines the test strategy, scope, schedule, and responsibilities for validating the Web Based ML Model Evaluator v1.0, covering the objectives, scope, strategy, resources, schedule, and responsibilities for verifying data ingestion, data preprocessing, machine learning task routing, model execution, performance evaluation, visualization, side-by-side comparison, and non-functional requirements.

Scope: Testing covers end-to-end browser features including CSV uploads (<10 MB), preview/summary, missing value handling, task routing (Classification, Regression, Clustering), parameter configuration, algorithm execution, visualizations (Confusion Matrix, 2D PCA plots), comparison dashboard, error handling, and security/performance constraints. Excludes host hardware and internal logic of third-party libraries (e.g., scikit-learn, pandas).

References:
- Software Requirements Specification (SRS) for Web Based ML Model Evaluator v1.0 (Group 13)
- Functional Requirements Document - Web Based ML Model Evaluator (Team 13)
- IEEE Recommended Practice for Software Requirements Specifications

Definitions:
- **ML:** Machine Learning
- **SRS:** Software Requirements Specification
- **CSV:** Comma Separated Values
- **PCA:** Principal Component Analysis
- **KNN:** K-Nearest Neighbors
- **MAE / MSE / RMSE:** Mean Absolute Error / Mean Squared Error / Root Mean Squared Error
- **RTM:** Requirements Traceability Matrix
- **UAT:** User Acceptance Testing

2. Test Items

1. Data ingestion and preprocessing module
- Includes file upload, file preview and basic missing value handling.

2. ML task routing and configuration module
- Includes selecting the problem type (classification, regression and clustering), target selection, cluster count inputs, train/test splits inputs (if the problem type requires it).

3. Algorithm execution and metric computation module
- Execute the problem type and calculate the metrics.

4. Model comparison and visualization module
- Side by side comparison of multiple trained models, display of confusion matrix, plots and calculated metrics.

5. Error handling
- Error handling will be a separate component in the project.
- Checking invalid file formats, ensuring size limits, etc and structure and send the error message to the UI.

3. Features to be Tested

- FR-2: 5 row data preview of uploaded csv file
- FR-3: Preprocessing and missing value handling
- FR-4: ML Task Selection such as classification, regression, clustering
- FR-5: Feature and target selection for supervised learning tasks
- FR-6: Feature selection and cluster count configuration for clustering specific problem
- FR-7: Train/test data partitioning from user
- FR-8 / BR-5: Task specific algorithm execution

  Eg:

  Classification: Logistic regression, decision tree, KNN, etc

  Regression: Linear regression, decision tree regressor, etc

  Clustering: K-Means clustering

- FR-9 / BR-6: Classification metrics & visualizations (accuracy, precision, recall, F1 score, confusion matrix)
- FR-10 / BR-6: Regression metrics (MAE, MSE, RMSE, R^2 score)
- FR-11 / BR-6: Clustering metrics & visualizations (silhouette score, davies-bouldin Index, PCA scatter plot)
- FR-12 / BR-7: Model comparison dashboard having side by side metrics comparison
- FR-2 / SR-3 / SEC4: Error handling such as unsupported file types, unsupported algorithm specific user inputs and exceeding file size limit.

4. Features Not to be Tested

- Internal logic of underlying standard python libraries (scikit, numpy etc.)
- Data formats other than csv (non-csv upload rejection is tested under input validation)
- Physical web server hardware maintenance, host OS security, and cloud data center network infrastructure.

- Long term external database persistence (application uses session based in memory processing)

5. Test Approach / Strategy

Levels:
- **Unit Testing:** Validating individual backend functions (CSV parsing, pre-processing functions, metric calculation scripts, route handlers)
- **Integration Testing:** Testing API interaction between UI frontend and FastAPI/Python backend pipeline.
- **System Testing:** End-to-end evaluation from CSV upload to preprocessing to model configuration to execution to visualization and side-by-side comparison
- **Acceptance Testing (UAT):** Verifying the workflow simplicity and evaluation of metric correctness against business rules with domain user scenarios.

Types:
- **Functional Testing:** Verifying all REQ-1 through REQ-13 and FR-1 through FR-12 capabilities
-**Boundary & Negative Testing:** Uploading files exceeding 10 MB, invalid CSV structures, missing columns, selecting same target as feature, non-numeric regression targets.
- **Performance Testing:** Measuring execution times to ensure <30 second latency for <10 MB datasets.
- **Usability & Compatibility Testing:** Ensuring guided 4-step workflow rendering across Chrome, Firefox, and Edge .

- **Security & Memory Testing:** Verifying dataset protection, input sanitization, and session buffer memory purge.

Entry Criteria:

- Stable web application build deployed in test environment

- Test CSV datasets (Classification, Regression, Clustering scenarios) prepared.

- SRS and Functional Specifications baselined.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate input file type checking (strictly enforcing .csv files).
- Enforce file size boundaries with a hard 10 MB restriction to prevent denial-of-service (DoS) or resource exhaustion.
- Validate input parameters to prevent injection attacks or invalid configuration payloads.
- Verify session buffer cleanup by purging host memory of uploaded CSVs and model artifacts upon session termination.
- Validate error messages to ensure backend traces do not expose sensitive host server details.

6. Test Environment

Hardware: Test Workstations (Intel i5/i7, 16GB RAM)
OS: Windows 11 / Linux Ubuntu.
Backend Environment: Python 3, FastAPI framework, pandas, scikit-learn, numpy
Tools:
- API Testing: Postman
- Performance Testing: Apache JMeter / Locust
- Defect Tracking & Management: Jira

Test Data: Standard benchmark CSV datasets under 10 MB, corrupted/malformed CSV files, zero-variance feature datasets, non-numeric regression datasets.

7. Test Schedule

Milestones:
- Test case design: 05-Oct-2025
- Environment setup: 07-Oct-2025
- Test execution start: 08-Oct-2025
- Test execution end: 20-Oct-2025
- UAT: 22-Sep-2025 to 25-Oct-2025

8. Test Deliverables

- Software Test Plan (this document)
- Test Cases Specification Document (Manual & Automated Scripts)

- Requirements Traceability Matrix (RTM)
- Test Data Sets (Valid & Synthetic Invalid CSVs)
- Execution Logs & Performance Metrics Reports
- Defect Reports (Jira export)
- Final Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Sanyam Deven Hiran	Prepare plan, coordinate execution
Test Engineer	Shreya Patil, Shreya Ajith, Satwik Phadka, Sanyam Deven Hiran	Design & execute test cases, log defects
Developer	Shreya Patil, Shreya Ajith, Satwik Phadka, Sanyam Deven Hiran	Support defect fixes and triage
Product Owner	Sudeepa Roy Dey	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	**Mitigation**
Processing delays exceeding the 30-second pipeline response latency limit for model training (NFR-1).	Perform early performance testing with datasets nearing the 10 MB limit and optimize backend model configurations.
File upload failures or server slowdowns due to excessively large datasets.	Enforce and test strict maximum file size restrictions (NFR-4) before processing begins.

Unstructured or corrupted CSV files breaking the preview (FR-2) or preprocessing (FR-3) modules.	Ensure robust error handling is tested to verify it intercepts invalid states and returns structured, human-readable error messages to the UI (FR-2).
Browser compatibility issues leading to UI rendering failures.	Execute cross-browser testing on Google Chrome, Firefox, and Microsoft Edge early in the test cycle (NFR-3).

11. Assumptions & Dependencies

- **Assumptions:**
  - Users will upload datasets in valid CSV formats suitable for analysis.
  - The user interface follows the defined clear process of upload, selecting columns, training the model, and displaying results (NFR-2).
  - The selected target variable and features provided by the user will be appropriate for the chosen ML problem type.
- **Dependencies:**
  - Availability of modern web browsers (Google Chrome, Firefox, Microsoft Edge) for UI validation without requiring extra plugins (NFR-3).
  - Availability of sample test datasets (under 10 MB) suitable for Classification, Regression, and Clustering tasks.
  - Stability of backend machine learning algorithms (e.g., Logistic Regression, Random Forest, K-Means, etc.) required for task-specific execution (FR-8).

12. Suspension & Resumption Criteria

- **Suspend testing if:**
  - The web application environment is unavailable for >4 hours.
  - Core functionality defects block essential pathways, such as the inability to upload a dataset (FR-1) or select an ML task (FR-4).
  - The system consistently crashes instead of returning structured error messages during invalid execution states (FR-2).
- **Resume testing if:**
  - Critical blocking defects (like file upload failures or dataset preview crashes) are resolved by the development team.
  - The test environment is stabilized and accessible.

13. Test Case Management & Traceability

Requirement Traceability Matrix (RTM):

Sl. No.	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	Pass/ Fail
1	FR-1	User should be able to upload their dataset file here.	AM-1: Dataset Ingestion and Preprocessing	SF-1: Data Ingestion and Preprocessing	TBd	T1	TBD
2	FR-2	System should analyze the uploaded CSV and present a 5 record dataset preview and show total rows, column names, data types, missing value	AM-1: Dataset Ingestion and Preprocessing	SF-1: Data Ingestion and Preprocessing	TBD	T2	TBD
3	FR-3	System should preprocess and clean the dataset	AM-1: Dataset Ingestion and Preprocessing	SF-1: Data Ingestion and Preprocessing	TBD	T3	TBD
4	FR-4	System should allow the user to select an ML problem type such as classification or regression	AM-2: ML Task Configuration	SF-2: ML Task Routing and Configuration	TBD	T4	TBD
5	FR-5	For classification and regression tasks, the system should allow the user to select one column as the target variable and one or more columns as features.	AM-2: ML Task Configuration	SF-2: ML Task Routing and Configuration	TBD	T5	TBD

6	FR-6	For clustering, the system should allow the user to input the number of clusters.	AM-2: ML Task Configuration	SF-2: ML Task Routing and Configuration	TBD	T6	TBD
7	FR-7	For supervised tasks, the system shall provide an option to adjust the train/test split ratio.	AM-2: ML Task Configuration	SF-2: ML Task Routing and Configuration	TBD	T7	TBD
8	FR-8	System should execute model fitting based on user selection-classification, regression and clustering algorithms.	AM-3: Model Training and Evaluation	SF-3: Algorithm Execution and Metric Computation	TBD	T8	TBD
9	FR-9	System should compute and display accuracy, precision, recall, F1 score, and a confusion matrix for classification.	AM-3: Model Training and Evaluation	SF-3: Algorithm Execution and Metric Computation	TBD	T9	TBD
10	FR-10	System should compute and display MAE, MSE, RMSE, and R^2 where applicable.	AM-3: Model Training and Evaluation	SF-3: Algorithm Execution and Metric Computation	TBD	T10	TBD
11	FR-11	System should compute Silhouette Score and Davies-Bouldin Index and display a 2D PCA scatter plot showing cluster assignments.	AM-3: Model Training and Evaluation	SF-3: Algorithm Execution and Metric Computation	TBD	T11	TBD
12	FR-12	System should allow users to compare evaluation metrics of two or more trained models side by side.	AM-4: Model Comparison	SF-4: Model Comparison Dashboard	TBD	T12	TBD
13	FR-13	Backend should intercept invalid execution states and return error	AM-1, AM-2, AM-3, AM-4	SF-4: Error Handling	TBD	T13	TBD

		messages to the UI.					
14	NFR-1	Ingestion, feature scaling, model fitting, and metric generation for datasets under 10 MB shall execute within 30 seconds per run.	Overall Architecture	Performance Requirements	TBD	T14	TBD
15	NFR-2	The web interface should guide the user through a clear process	AM-1, AM-2, AM-3, AM-4	UI / Usability Design	TBD	T15	TBD
16	NFR-3	The web application should run smoothly on Google Chrome, Firefox, and Microsoft Edge without requiring extra browser plugins.	Overall Architecture	External Interface / Browser Compatibility Design	TBD	T16	TBD
17	NFR-4	The system should restrict file uploads to a maximum size to prevent server slowdowns.	AM-1: Dataset Ingestion and Preprocessing	SF-1: Data Ingestion and Preprocessing	TBD	T17	TBD

14. Test Metrics & Reporting

Metrics:

- % Test Cases Executed vs. Planned
- % Pass / Fail Rate
- Defect Density (Defects per Functional Module)
- Defect Severity Breakdown (Critical, Major, Minor)
- Pipeline Execution Latency Timings (seconds per dataset)
- Requirement Coverage Percentage (Target: 100%)

Reports:

- Daily Test Execution Status Update
- Weekly Defect Tracking & Triage Summary
- Final Test Sign-Off Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Sanyam Deven Hiran	
Test Engineer	Shreya Patil, Shreya Ajith, Satwik Phadka, Sanyam Deven Hiran	
Developer	Shreya Patil, Shreya Ajith, Satwik Phadka, Sanyam Deven Hiran	
Product Owner	Sudeepa Roy Dey	